



रक्षा मंत्रालय
MINISTRY OF DEFENCE

ज्वाइंट सर्विस स्पेसिफिकेशन
JOINT SERVICES SPECIFICATION

ON

SAREES LIGHT BLUE
(DS Cat No. CK / 8410 - 000 059)

मानकीकरण निदेशालय
रक्षा उत्पादन विभाग
रक्षा मंत्रालय
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DIRECTORATE OF STANDARDISATION
DEPARTMENT OF DEFENCE PRODUCTION
MINISTRY OF DEFENCE
'H' - BLOCK, NIRMAN BHAVAN, P.O.
NEW DELHI - 110 011

LIST OF MEMBERS ASSOCIATED WITH FORMULATION OF THIS STANDARD

1. This Joint Services Specification has been approved by Shri Sanjeev Kapur Addl DGQA (S), Chairman, Stores Standardisation Sub - Committee.
2. The following members have been associated in preparing the document :

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(j)	Maj. S. Bansal, DD OS (PII)	MGO / CP Cell / OS Dte.
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RECORD OF AMENDMENTS

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0.0 FOREWORD

0.1 This Joint Services Specification has been prepared by Stores Standardisation Sub Committee on the authority of Standardisation Committee, Ministry of Defence.

0.2 This specification has been approved by the Ministry of Defence and is mandatory for use by the Defence Services.

0.3 This specification has been prepared on the basis of Specification Number IND / TC / 0020 (a) and supersedes the same.

0.4 This specification would be used for Manufacture, Quality Assurance and Procurement of the item.

0.5 Quality Assurance Authority for the item covered in this specification are CQA (T & C), Kanpur for Army, Navy and Air Force. Enquiries regarding technical parameters shall be addressed to the Quality Assurance Authority, while other enquiries shall be referred to :

The Director,
Directorate of Standardisation,
Ministry of Defence,
'H' Block, Nirman Bhavan PO,
New Delhi – 110 011

0.6 Non - registered users can obtain the following on payment :

(a) Copies of IS from :

Bureau of Indian Standards,
Manak Bhavan,
9, Bahadur Shah Zafar Marg,
New Delhi – 110 002
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their regional / branch offices

(b) Copies of JSS / JSG from :

The Director,
Directorate of Standardisation,
Standardisation Documents Centre
Ministry of Defence
Room No.05, 'J' Block, Nirman Bhavan PO,
New Delhi – 110 011

0.7 Indian Standard (IS) specifications are available free of cost for registered users on :

Directorate of Standardisation Website
www.ddpdos.gov.in
for registration visit our website.

0.8 This specification holds good only for the supply order for which it is issued.

0.9 **Directorate of Standardisation Website.** All the approved JSSs / JSGs are available on the Directorate of Standardisation Website **www.ddpdos.gov.in** Defence organisations desirous of accessing a copy of this document are requested to approach the Directorate of Standardisation for obtaining user id / password to access the website.

1.0 SCOPE

1.1 This specification requires of against Sarees Light Blue (DS Cat No. CK / 8410 - 000 059) defence requirements and provides guidance to contractors / suppliers, manufacturers, QA agencies and stockists / indentors etc.

1.2 This Specification covers the requirements and methods of test for Sarees Light Blue used for female patients in hospitals.

2.0 RELATED SPECIFICATIONS

2.1 Reference is made in this specification to :

Sl. No.	Specification / Document No.	Nomenclature
(a)	IS 171 : 1993 (Fourth Revision) Reaffirmed - 2007 Amd - 1	Textiles - Ring spun grey cotton yarn for weaving - Specification
(b)	IS / ISO 105 - C 10 : 2006	Textiles – Tests for Colour Fastness : Part C10 : Colour Fastness to Washing with Soap or Soap and Soda
(c)	IS 1029 : 1970 (First Revision) Reaffirmed - 2013 Amd - 3	Specification for Hot Rolled Steel Strips (Baling)
(d)	IS 1315 : 1977 (First Revision) Reaffirmed - 2010 Amd - 1	Method for determination of linear density of yarns spun on cotton system
(e)	IS 1383 : 1977 (First Revision) Reaffirmed - 2009	Method for determination of scouring loss in grey and finished cotton textile materials
(f)	IS 1390 : 1983 (First Revision) Reaffirmed - 2009	Methods for determination of pH value of aqueous extracts of textile materials
(g)	IS 1912 : 1984 (Second Revision) Reaffirmed - 2011 Amd - 1	Specification for Country Jute Twine
(h)	IS 1954 : 1990 (Second Revision) Reaffirmed - 2007	Determination of Length and Width of Woven Fabrics - Methods

Sl. No.	Specification / Document No.	Nomenclature
(j)	IS 1963 : 2004 (Second Revision) Reaffirmed - 2008	Methods for Determination of Threads per Unit Length in Woven Fabrics
(k)	IS 1964 : 2001 (Second Revision) Reaffirmed - 2010	Textiles - Methods for determination of Mass per unit length and mass per unit area of fabrics
(l)	IS 1969 : 1985 (Second Revision) Reaffirmed - 2010 Amd - 1	Methods for Determination of Breaking load and Elongation of Woven Textile Fabrics
(m)	IS 2454 : 1985 (First Revision) Reaffirmed - 2013 Amd - 1	Methods for determination colour fastness of textile materials to artificial light (xenon lamp)
(n)	IS 2508 : 1984 (Second Revision) Reaffirmed - 2008 Amd - 3	Specification for Low Density polyethylene Films
(o)	IS 2977 : 1989 (First Revision) Reaffirmed - 2010 Amd - 1	Fabrics (other than wool) - Method for Determination of dimensional changes on soaking in water
(p)	IS 4905 : 1968 Reaffirmed - 2011 Amd - 1	Methods for random sampling
(q)	IS 6117: 1977 (First Revision) Reaffirmed - 2012 Amd - 2	Specification for - Tapes, Cotton
(r)	ISS 8135 - 10 : 2004 (Revision No. 1) Reaffirmed - 2011 Amd - 1	Polypropylene Strapping, 12 mm wide
(s)	IND / TC / 2123 (b)	Laminated Cloth Hessian

* All the IS specification updated based on BIS Website / handbook 2013

- 2.2 Copies of IND / TC specification are obtainable on payment from :

The Controller,
Controllerate of Quality Assurance (Textiles & Clothing),
Post Box No. 294,
Kanpur - 208 004 (U.P.)

- 2.3 Copies of CIGS / US are obtainable on payment from :

The Controller
Controllerate of Quality Assurance (General Stores)
Post Box No. 127
Kanpur - 208 004 (U.P.)

3.0 STANDARD PATTERN

- 3.1 Standard pattern of Sarees Light Blue (DS Cat No. CK / 8410 - 000 059) held in the custody of Controllerate of Quality Assurance (Textiles & Clothing), Kanpur shall constitute the standards as regards appearance, shade, workmanship, feel, finish and any other particulars or properties not noted defined in this specification.

4.0 MATERIALS

- 4.1 The Sarees shall be manufactured from Gray Cotton Yarn conforming to IS 171 : 1993

5.0 MANUFACTURE

- 5.1 Sarees shall be dyed Light Blue of required shade and should be free from all defects like patchiness, streaky effect etc.

- 5.2 The fabric shall be free from substances like starch and filling material, which are liable to cause damage to the cloth during storage.

- 5.3 The sarees shall be free from spinning and weaving defects like slubs, neps, bad piecing etc.

6.0 REQUIREMENTS

- 6.1 Sarees Light Blue shall conform to the following particulars :

Sl. No.	Characteristics	Requirements	Methods
(a)	Length	5.5 m + 25 cm - 15 cm	IS 1954 : 1990
(b)	Width	120 ± 25 cms	IS 1954 : 1990

Sl. No.	Characteristics	Requirements	Methods
(c)	Weight / m ²	40 g min	IS 1964 : 2001
(d)	Linear Density of Yarn in the body (suggested) Warp Weft	7.5 (Tex) 7.5 (Tex)	IS 1315 : 1977
(e)	Threads / dm in the body of the Sarees Warp Weft	280 ± 5 % 280 ± 7.5 %	IS 1963 : 2004
(f)	Minimum breaking load in the body of the sarees (5 x 20 cm between the grips :) Warp Weft	130 N 130 N	IS 1969 : 1985
(g)	Weave	Plain	
(h)	Width of selvedge	1 cm	(2 fold warp yarn to be used in the selvedge)
(j)	Scouring loss percent (max.) for Power loom for Hand loom	5 % 7 %	IS 1383 : 1977 (Mild method)
(k)	Maximum Dimensional changes due to shrinkage or elongation in warp and weft direction	5 %	IS 2977 : 1989
(l)	pH value of aqueous Extract	6.0 to 9.0	IS 1390 : 1983 (Cold method)

Sl. No.	Characteristics	Requirements	Methods
(m)	Colour fastness to : (i) Light	(aa) Class V or better or (ab) Rating 5 or better	(aa) Classification adopted by Society of Dyers & Colorist, Bradford, Yorkshire, when exposed to light from high intensity carbon arc lamp in fadometer (ab) IS 2454 : 1985
	(ii) Washing	Rating 4 for change in shade and for bleeding on the attached piece of white cotton material.	IS / ISO 105 - C 10 : 2006

7.0 MARKING

7.1 Each saree prior to being offered for Quality Assurance shall be legibly and indelibly marked at one end with Cat. No, Nomenclature of the store, length of the Saree in metres, month and year of manufacture and Contractor Name, initials or his recognised trade mark.

8.0 ACCEPTANCE MARK

8.1 Acceptance mark be stamped within 10 to 15 cm of firms marking with a A / mark of 25 mm or 13 mm size using indelible-ink / paint.

9.0 PRE - INSPECTION OF STORES / CONSIGNMENT

9.1 Manufacturers / contractors must satisfy themselves that the stores are in accordance with the terms of the contract and fully conform to the required specification by carrying out a thorough pre - inspection of each lot before actually tendering the same for inspection to the Quality Assurance Officer nominated under the terms of the contract. A declaration by the contractor that necessary pre - inspection has been carried out on the stores tendered, will be submitted alongwith the challan. The declaration will also indicate the method followed in carrying out pre - inspection showing the features checked / tested and will have the test certificate attached to the challan / declaration.

9.2 If the Quality Assurance Officer finds that pre - inspection of the consignments as required above has not been carried out, the consignment is liable for rejection.

10.0 SAMPLING PROCEDURESAMPLING PLAN FOR MADE UP STOREBASED ON AQL of 4 %

AUTHORITY ISO : 2859 - 1989 / IS 2500 : Part - I : 2000

Sl. No.	Lot Size in Nos	Sampling Plan for					
		Visual Examination / Dimensional check at the time of sampling (Gen. Insp. Level - I)		Physical Parameters (for laboratory tests) (Spl Insp Level S - 3)		Chemical Parameters and other physical parameters requiring long testing time (for laboratory tests) (Gen. Insp. Level - II)	
		Sample Size (3)	Acceptance No. (4)	Sample Size (5)	Acceptance No. (6)	Sample Size (7)	Acceptance No. (8)
(1)	(2)						
(a)	Below - 280	13	1	8	1	3	0
(b)	281 - 500	20	2	8	1	5	0
(c)	501 - 1200	32	3	13	1	8	1
(d)	1201 - 3200	50	5	13	1	13	1
(e)	3201 - 10000	80	7	20	2	20	2

NOTE : (1) Sampling shall be done randomly as per IS 4905 : 1968.

(2) Sample drawn as per column (3) to be first examined visually. If found satisfactory the sample for lab testing as per column (5) be drawn out from the sample originally drawn as per column (3) by the sampling officer and marked by him accordingly.

(3) For chemical parameters sample to be drawn at random as per IS 4905 : 1968 from those samples drawn as per column (5).

(4) Lot size 10,000 Numbers (max).

11.0 INSPECTION

11.1 If, On examination of samples taken from any portion of the consignment, the material is found to be not fully in accordance with the specification, the whole supply may be rejected.

12.0 WARRANTY

12.1 Except as otherwise provided in the invitation to the tender, the contractor / seller will declare that the goods, stores, articles sold / supplied to the purchaser under this contract shall be of the best quality and workmanship and new in all respects and shall be strictly in accordance with the specifications and particulars contained / mentioned in the contract.

12.2 The contractor / seller hereby guarantees that the said goods / stores articles would continue to conform to the description and quality aforesaid for a period of twelve months from the date of delivery of the said goods / stores / articles to the purchaser or 15 months from the date of shipment / dispatch from the contractor's work, whichever is earlier and that notwithstanding the fact that the purchaser (Inspector) may have inspected and / or approved the said goods / stores / articles, if during the aforesaid period of 12 / 15 months the said goods / stores / articles, be discovered not to be conform to the description and quality aforesaid or not having satisfactory performance or have deteriorated and the decision of the purchaser in that behalf shall be final and binding on the contractor / seller to rectify / replace by acceptable goods / stores / articles, or such portion or portion thereof as is found to be defective by the purchaser within a reasonable period not exceeding 3 months or as may be allowed by the purchaser in his discretion on the application made thereof by the contractor / seller and in such an event the above mentioned warranty period shall apply to the goods / stores / articles rectified / replaced from the date of rectification / replacement thereof, otherwise the contractor / seller shall pay to the purchaser such compensations as determined by the purchaser as may arise by reason of the break of the warranty herein contained.

13.0 PACKAGING

13.1 Materials - The following material shall be used for packing of Sarees :

Sl. No.	Material	Conforming to
(a)	Polyethylene film 0.04mm thick	IS 2508 : 1984
(b)	Laminated Cloth Hessian	IND / TC / 2123 (b)
(c)	Twine Jute 3 ply	IS 1912 : 1984
(d)	Mild Steel hoops 16 mm wide x 0.9 mm thick or Polypropylene Strap	IS 1029 : 1970 (Med. Grade)
(e)	Slat wood	Best Trade Quality
γ	Tape Cotton 13 mm	IS 6117 : 1977

13.2 Method

13.2.1 The stores shall be supplied in new, clean, dry and sound condition.

13.2.2 Each Saree shall be tied with tape cotton 13 mm width at two places to keep its folding intact and packed in polythene bag made from polythene film 0.04 mm thick.

13.2.3 100 such Sarees shall then be wrapped with an inner and outer layer of laminated Hessian Cloth to form a compact bale of rectangular shape. The overlaps of inner layers shall be at least 10 cm to ensure full protection of the content of the bale. The overlap of the outer layer shall be such that it can be securely sewn around the bale. The polyethylenene face of the laminated hessian cloth shall be kept on inner side of the bale. The bale shall be stitched with double 3 ply jute twine with not less than 6 stitches per 10 cm taking care not to pierce the inner strapping during stitching. Sufficient hessian cloth shall be pulled out at each corner to form ears of minimum 15 cm in length to facilitate easy handling during transit.

13.2.4 Slats of wood approximately 7 cm wide and 1 cm thick and free from rot or other harmful fungal decay or insect attack alternatively, bamboo strips duly preserved, shall be place length wise along the edges of the bale and finally bound with at least two bands of polypropylene strapping / metal hoops. Each strapping / hoop shall be one complete piece and the ends shall be sealed with machine.

13.2.5 The bales shall be bound with additional bands of polypropylene strapping / metal hoops where the dimensions of the bale may exceed 60 cm in any of the directions, so that no bands / hoops are more than 25 cm apart.

13.2.6 In all cases, gross mass of the bale shall not exceed 40 kg.

13.2.7 The seams of the bales shall be dammered with indelible ink / paint.

NOTE : The bale shall be fully pressed and made compact.

13.3 Marking

13.3.1 Before dispatch each bale shall be legibly marked by stencil with indelible marking ink / paint, showing the following details :

- (a) D. S. Catalogue Number and nomenclature of the store.
- (b) Quantity packed in the bale.
- (c) Lot and Serial number of the bale.
- (d) Month and year of packing.
- (e) Gross mass of the bale.
- (f) Name and address of the consignee.
- (g) Name / trade mark of the consignee.
- (h) Acceptance of Tender, Supply Order or Ext Number and Date.
- (j) Inspection / Note Number & Date.

14.0 DEFENCE STORES CATALOGUE NUMBER

14.1 The store covered by this specification shall bear the following DS Cat Number :

Sl. No	DS Cat No	Nomenclature
(a)	CK / 8410 - 000 059	Sarees Light Blue

15.0 SUGGESTIONS FOR IMPROVEMENT

15.1 Any suggestion for improvement in this document may be forwarded to :

The Director,
Directorate of Standardisation,
Ministry of Defence,
'H' Block, Nirman Bhavan PO,
New Delhi - 110 011

1000

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क्रम संख्या 52407 सामान्य/प्राथमिक
विभाग के अनुसार रक्षा
मंत्रालय द्वारा अनुमोदित, हस्ताक्षर से
.../.../... को है। SSSC
Approved by Ministry of Defence
Vide Sr. No.
CoP/Std dated ... 31/9/19
This document contains pages



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भारत सरकार
GOVERNMENT OF INDIA
रक्षा मंत्रालय
MINISTRY OF DEFENCE
संयुक्त सेवा विनिर्देश
JOINT SERVICES SPECIFICATION

ON

SHEET (VARIOUS)
(DS Cat Part No. 7210-000 024, 7210-000 026 & 7210-000 049)
(NSN 7210720475216, 7210720099866 & 7210720475215)

मानकीकरण निदेशालय
रक्षा उत्पादन विभाग, रक्षा मंत्रालय
'एच' ब्लॉक, निर्माण भवन डाकघर
नई दिल्ली - ११००११

DIRECTORATE OF STANDARDISATION
DEPARTMENT OF DEFENCE PRODUCTION
MINISTRY OF DEFENCE
'H' BLOCK, NIRMAN BHAWAN PO
NEW DELHI - 110011

LIST OF MEMBERS ASSOCIATED WITH FORMULATION OF THIS STANDARD

1. The third revision of this Joint Services Specification 7210-25 has been approved by Shri P Meena, ADGQA (S), Chairman, Stores Standardisation Sub Committee by circulation.
2. The following members were consulted in approving the document:

S. No.	Name & Designation	Organisation
1.	Gp Capt R P Dubey, JDQAS	DQAS(Aero)/Air HQ
2.	Shri VR Choudhary, PScO	CQA (T&C)
3.	Dr Gurubachan Singh, JAG (NFSG)	CQA (M)
4.	Dr OP Singh, SSO-I	CQA (PP)
5.	Er. M. Satyanarayana	CQA (GS)
6.	Cdr Naveen Kumar (IN)	DQA (N)
7.	Shri Jarnail Singh, Dy Director	DGAQA
8.	Maj S Bansal, DD OS (PII)	MGO/CP Cell/OS Dte.
9.	Lt Col Ashish Kumar, JDOS	MGO Br/EM/GS & C
10.	Wg Cdr Chaturvedi, Dy Dir	Dte of Stores/Air HQ
11.	Lt Cdr KNVS Ganesh Karthik	DDNA
12.	Lt Cdr Pawan Deep	NHQ/DCV
13.	Cdr Rohit Prakash, JDME	NHQ/DOS (E)
14.	Dr DN Tripathi, Sc 'F'	DMSRDE
15.	Lt Cdr Pawan Singh, JDLS	JDLS
16.	Shri S K Bandopadhyay, JGM	OEF, Kanpur
17.	Lt Col Pradeep Thapa	Secy, SSSC

सहायक निदेशक (तक सम)
Asst Director (TC)
मानकीकरण निदेशालय
Directorate of Standardisation
RECORD OF AMENDMENTS

Amendment		Amendment pertains to S. No./Para No./ Column No.	Authority	Amended by	Signature & Date
No.	Date			Name & Appointment (In block letters)	

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0. FOREWORD

0.1 This specification has been prepared by the Stores Standardisation Sub Committee on the authority of Standardisation Committee, Ministry of Defence.

0.2 This specification has been approved by the Ministry of Defence and is mandatory for use by the Defence Services.

0.3 This JSS 7210-25 : 2019 (Third Revision):

- a) was revised in the year 2008 and 2013.
- b) is revision of JSS 7210-25 : 2013 (Second Revision) and supersedes the same.

0.4 This specification would be used to Guide, Manufacture, Quality Assurance and Procurement of the item.

0.5 In the said specification following changes have been made with respect to JSS 7210-25 : 2013 (Second Revision):

- a) NSN has been added.
- b) Related specifications have been updated.
- c) Clauses pertaining to Quality Assurance and Sampling, have been updated as per current policies.

0.6 Quality Assurance Authorities for the item covered in this specification are CQA (T&C), Kanpur-208004, Email address of the CQA (T&C) Kanpur is cqatcknp-dgqa@nic.in. Enquiries regarding any technical parameter shall be addressed to the Quality Assurance Authority, while other enquiries shall be referred to Secy SSSC by email at sssc.defstand@gov.in or by post to:

The Director,
Directorate of Standardisation,
Ministry of Defence,
'H' Block, Nirman Bhawan PO,
New Delhi-110011.

0.7 Non-registered users can obtain the following on payment:

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समानक निदेशक (सक सम)
Asstt. Director (TC)
समानिकरण निदेशालय
Directorate of Standardisation
एम. संवालय, नई दिल्ली-110011
Ministry of Defence, New Delhi-110011

1. SCOPE

This specification covers the requirements of various sizes of sheets fabricated from cotton sheetings. (DS Cat Part No. 7210-000 049, 7210-000 024, 7210-000 026 and NSN 7210720475215, 7210720475216 & 7210720099866).

2. RELATED SPECIFICATIONS

References are made in this specification to:

Table 1 Related Specifications

S. No.	Specification No.	Nomenclature
a)	IS 175 : 1989 (Third Revision) Amd 1 Reaffirmed 2017	Sheeting, Tickings and Bedsheets-Specification
b)	IS 1720 : 1978 (Second Revision) Reaffirmed 2017	Specification for Cotton Sewing Threads
c)	IS 2454 : 1985 (First Revision) Amd 1 Reaffirmed 2017	Method for Determination of Colour Fastness of Textile Materials to Artificial Light (Xenon Lamp)
d)	IS 2508 : 2016 (Third Revision)	Polythene Film and Sheets-Specification
e)	IS 2771 (Part 1) : 1990 (Second Revision) Amd 3 Reaffirmed 2018	Corrugated Fibreboard Boxes-Specification Part 1 General Requirements
f)	IS 4905 : 2015/ ISO 24153 : 2009 (First Revision)	Random Sampling and Randomization Procedures
g)	IS 9738 : 2003 (Second Revision) Reaffirmed 2015	Polyethylene Bags for General Purposes-Specification
h)	IS 13262 : 1992 Amd 1 Reaffirmed 2018	Pressure Sensitive Adhesive Tapes with Plastic Base-Specification
j)	JSS 8135-10 : 2016 (Second Revision)	Polypropylene Strapping, 0.55 mm thick x 12 mm wide
k)	IS/ISO 105-C10 : 2006	Method for determination of colour fastness to washing test B-2
m)	Label cardboard 45 mm x 25 mm with-tie-on-tag	Best Trade quality to be approved by the Quality Assurance Officer

3. STANDARD PATTERN

The Standard pattern held by The Controller, Controllerate of Quality Assurance (Textiles & Clothing), Ashok Path, Kanpur-208004 shall constitute the standard as regards any particulars or properties not noted or defined in this specification.

4. MATERIAL

The sheets shall be manufactured from the following materials conforming to the particulars noted against each.

Table 2 Material

S. No.	Components	Materials	Particulars
a)	Body	Sheeting Cotton bleached 910 mm	IS 175 (Variety No. 2)
b)	Sewin, Thread	Sewing Cotton, 9.8 Tex/6. white (bleached)	IS 1720

5. MANUFACTURE

5.1 The sheets shall be fabricated to the required dimensions as shown in clause 6 below. The two sides shall have selvages of the body material and the two ends shall be hemmed 19 mm with a turn in of 6 mm at each end. The stitching shall be of even tension throughout with all loose ends securely fastened off. The number of stitches shall be 11 to 13 per 30 mm.

5.2 Sheet Hospital DS Cat Part No. 7210-000 026 shall have the following additional requirement.

- a) Bed sheet shall have 4 cm wide blue strip woven along the warp direction on both side 5 cm away from the selvages. Blue shade shall be as per standard held in the custody of the Controller CQA(T&C), Kanpur-208004.
- b) Fastness to light of blue strip when exposed the high intensity carbon arc shall be class 4 (four) or better.
- c) Fastness to washing to blue strip shall be class (four) or better.
- d) The stipulation of residual starch content is not applicable.

6. DIMENSIONS AND TOLERANCE

The sheets shall conform to the following dimension:

Description with DS Cat Part No. & NSN	Dimensions	Width	Width of hem
	Length including hem		
Sheet Baby 7210-000 049 7210720475215	1.3 m $\pm$ 20 mm	910 mm $\pm$ 10 mm	20 mm
Sheet Barrack 7210-000 024 7210720475216	2.40 m $\pm$ 20 mm	1350 mm $\pm$ 10 mm	20 mm
Sheet Hospital 7210-000 026 7210720099866	2.55 m $\pm$ 20 mm	1750 mm $\pm$ 10 mm	20 mm

7. WORKMANSHIP AND FINISH

7.1 Sheets Baby and Barrack shall have a scoured finish free from sizing and filling materials.

7.2 Sheet Hospital shall have a full bleached finish, free from blueing, sizing, filling and other dressing agents.

7.3 Sheet shall be free from substances liable to cause subsequent tendering.

7.4 In appearance, general workmanship and finish, and in all other respects, not noted/defined in this specification, the sheets shall conform to the corresponding Standard Pattern.

8. MARKING ON STORE

Each sheet, prior to being offered for inspection shall have a white/tuflon cloth label size 50 mm x 40 mm securely sewn at both end of corner at inside the cloth on which the supplier shall legibly marked with indelible ink including following particulars:

- Nomenclature of the Store.
- DS Cat Part No., NSN.
- Manufacturer's name, initials or recognised trade mark.
- Month and year of manufacture.
- Length and Width of sheet (Size).

9. PRE-INSPECTION OF STORES/CONSIGNMENT

9.1 Manufacturers/Contractors must satisfy themselves that the stores are in accordance with the terms and fully conform to the required specification by carrying out a thorough pre-inspection of each lot before actually tendering the same for Quality Assurance Officer nominated under the terms of the contract. A declaration by the contractor that the necessary

pre-inspection has been carried out on the stores tendered, will be submitted alongwith the challan. The declaration will also indicate the method followed in carrying out pre-inspection showing the features checked/tested and will have the test certificate attached to the challan/declaration.

9.2 If the Quality Assurance Officer finds that pre-inspection of the consignment as required above has not been carried out, the consignment is liable for rejection.

9.3 The store shall be offered in bundle of 5 (five) pieces.

10. QUALITY ASSURANCE

10.1 On receipt of supply order the Quality Assurance Process comprises of the following (whenever/wherever applicable):

- a) Assessment of firm's preparedness to execute the order.
- b) Visual Inspection, sampling, lab testing, verdicting, packing and confirmation followed by issue of I/Note.
- c) Quality Audit (both product & process) and surveillance.
- d) Surprise checks, vigilance checks and consignee end checks.
- e) Periodical monitoring of product both in storage and exploitation.
- f) Addressing defect reports, measures and fixing accountability
- g) Collecting, collating presenting & inferring the feedback date for specification amendments and product improvements.
- h) Working out quality rating and supply rating to access the feasibility of self Certification.

10.2 Issue of Quality Improvements Notes to facilitate the manufacturer to enhance process efficiency and flow between the process. (Learn Process).

11. SAMPLING

11.1 **Formation of Lot** - The delivery shall be visually inspected by Quality Assurance Officer on the spot in the first instance to ascertain its homogeneity in respect of nature, source, year of manufacture, batch, uniformity of production etc. If the product units are homogeneous, the delivery shall be treated as one lot. If not, the product units shall be segregated by the supplier into separate groups, so that each group is homogeneous within itself to form a sub lot. The supplier shall arrange the unit of homogeneous lot in such a way that all the product units are easily accessible to the Sampling Officer from all sides to enable him to draw samples at random from any portion of the homogeneous lot. The sampling of the lot to be done as per table given below:

संशोधन विभाग (एक सम)
परीक्षण विभाग (TC)
एन.डी.ए.ए. विभाग
परीक्षण विभाग
परीक्षण विभाग
परीक्षण विभाग

Table 3 Authority ISO 2859-1:1999-ISO 2859-1 (Part 1) : 2000

Lot Size in No.	Sampling Plan for							
	Visual Examination/ Dimensional Check at the Time of Sampling		Physical Parameters (for Laboratory Tests)		Chemical Parameters and other Physical Parameters Requiring Long Testing Time (for Laboratory Tests)		Sample Size for Visual Check at Bulk QA Stage	
	Gen Insp. Level-2		Gen Insp Level (S-4)		Spl Insp Level (S-2)		Gen Insp Level-3	
	Sample Size	Acceptance No.	Sample Size	Acceptance No.	Sample Size	Acceptance No.	Sample Size	Acceptance No.
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Below 280	32	3	13	1	5	0	50	5
281 to 500	50	5	13	1	5	0	80	7
501 to 1200	80	7	20	2	5	0	125	10
1201 to 3200	125	10	32	3	8	1	200	14
3201 to 10000	200	14	32	3	8	1	315	21

NOTES

1. Samples drawn as per column (2) to be first examined visually.
2. If above is found satisfactory, the samples for lab testing as per column (4) be drawn out from the samples originally drawn as per column (2) by the Sampling Officer and marked by him accordingly.
3. For chemical parameters samples to be drawn at random as per IS 4905/ISO 24153-2009 from those samples drawn as per column (4).
4. Level-3 of Sampling Plan is applied to have enhanced visual discrimination in bulk.
5. Lot Size 10000 No. (Max).

12. CRITERIA FOR CONFORMITY

The lot shall be considered to be in conformity with the required standard if samples drawn for lab test as well as visual parameters are found satisfactory as per sampling plan.

13. WARRANTY

13.1 Except as otherwise provided in the invitation to the tender, the contractor/seller hereby declares that the goods, store, article sold/supplied to the purchaser under this contract shall be of the best quality and workmanship and new in all respect and shall be strictly in accordance with the specification and particulars contained/mentioned in the contract.

13.2 The contractor/seller hereby guarantee that the said goods/stores/articles would continue to conform to the description and quality aforesaid for period of twelve months from the date of delivery of the said goods/stores/articles to the purchaser or 15 months from the

date of shipment/dispatch from the contractors works whichever is earlier and that notwithstanding the fact that the purchaser (Inspector) may have inspected and/or approved the said goods/stores/articles, if during the aforesaid period of 12/15 months the goods/stores/articles, be discovered not to conform to the description and quality aforesaid or not giving satisfactory performance or have deteriorated and the decision of the purchaser in that behalf shall be final and binding on the contractor/seller to rectify/replace by acceptable goods/stores/articles, or such portion thereof as is found to be defective by the purchaser within a reasonable period not exceeding 3 month or as may be allowed by the purchaser in his discretion on the application made thereof by the contractor/seller, and in such an event the above mentioned warranty period shall apply to the goods/stores/articles rectified/replaced from the date of rectification/replacement thereof, otherwise the contractor/seller shall pay to the purchaser such compensations as determined by the purchaser as may arise by reason of the breach of the warranty herein contained.

14. PACKAGING

14.1 The following materials shall be used for packing of sheet (various):

Table 4 Packaging

S. No.	Materials	Conforming to
a)	Polythene film 0.04 mm thick	IS 2508
b)	Double walled corrugated fibreboard boxes. 5 ply (outer liner shall be laminated with polyethylene to provide water-proofness)	IS 2771 (Part 1)
c)	Polyethylene bags for general purposes	IS 9738
d)	Pressure sensitive adhesive tapes with plastic base	IS 13262
e)	Polypropylene strapping 12 mm wide	JSS 8135-10
f)	Label cardboard 45 mm x 25 mm with-tie-on-tag	Best Trade quality to be approved by the Quality Assurance Officer

14.2 Method

14.2.1 The stores shall be supplied in new, clean, dry and sound condition.

14.2.2 Each sheet shall be suitably folded and kept in a polythene bag. 10 such sheets (kept in polythene bag) shall be tied by a suitable twine cord to form a bundle, suitable No. of such bundles of one size shall be placed in a double walled corrugated fibre board box of suitable size previously lined with polythene film.

14.2.3 The flaps of the boxes shall be closed with pressure sensitive adhesive tape with plastic base and numero seal of the smallest convenient size shall be affixed fully covering the joints.

14.2.4 The boxes packed above shall be bound with two bands of polypropylene strapping in each direction along length and width. Each strapping shall be one complete piece and the ends shall be sealed with machine using steel seals of suitable size.

14.2.5 The gross mass of the box shall not exceed 40 kg.

14.3 Marking

Before dispatch the, boxes shall be legibly indelible marked as under:

- DS Cat Part No., NSN and nomenclature of the store.
- Quantity packed preceded by abbreviation 'Qty'.
- Month and year of packing.
- Name and address of the consignee.
- Contract no. and date.
- Mass of the package in kg.
- Number of individual package and total number of packages in the consignment eg. 1 of 4, 2 of 4, 3 of 4 and 4 of 4 when the consignment consists of 4 packages.
- Inspections Note number and date.

15. DEFENCE STORES CATALOGUE NUMBERS/NATO STOCK NUMBERS

The stores covered by this specification shall bear the following DS Cat Part No. & NSN as given against each:

S. No.	DS Cat Part No.	NSN	Nomenclature
a)	7210-000 049	7210720475215	Sheet Baby
b)	7210-000 024	7210720475216	Sheet Barrack
c)	7210-000 026	7210720099866	Sheet Hospital

16. SUGGESTIONS FOR IMPROVEMENT

Any suggestions for improvement to this document may be forwarded to:

The Director,
Directorate of Standardisation,
Ministry of Defence,
'H' Block, Nirman Bhavan PO,
New Delhi-110011.

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C-104

PAGE NAME TC4554 PITCH-12 MARGIN-75 FLP NO.W-1

GOVERNMENT OF INDIA, MINISTRY OF DEFENCE
DETA ORGANISATION

COS (SECTION K-3)

SPECIFICATION NO. DD/TC/4554(c)

NO. OF DRAWING PLATES ATTACHED-1

SUPERSEDES:

REAFFIRMED ON:

STORE CASE NO. S/3286/TC-21

SPECIFICATION FOR

SHEETS HOSPITAL DRAW 182 X 91cm

K-3/7210-000025

This specification consists of four pages.

This specification is the property of Government of India. This specification or any drawings, pattern or other information issued in connection therewith may only be used for a specific order placed by competent authority. It is not to be used for any other purpose whatsoever without the expressed written sanction of the Director General of Quality Assurance or his representatives.

All clauses in this specification shall be complied with in every respect irrespective of the source of supply of the material and/or components.

Should any discrepancy be found between this specification and sample or pattern loaned for any purpose, this specification and/or other connected specification or drawing shall be taken as correct.

ISSUED BY

CONTROLLER
CONTROLLERATE OF QUALITY ASSURANCE
TEXTILES & CLOTHING, P.B. NO. 294
KANPUR-208 004

Price: A 55.00

1. SCOPE

1.1 This specification covers the requirements and method of tests of Sheets Hospital Draw 182cm x 91cm used as bed sheets in hospitals.

2. RELATED SPECIFICATIONS

2.1 Reference is made in this specification to:-

- | | |
|------------------|--|
| (a) IS:138-1961 | Ready mixed paint marking for packages and petrol containers. |
| (b) IS:852-1964 | Animal glue for general wood working purpose. |
| (c) IS:1029-1970 | Hot rolled steel strips (baling). |
| (d) IS:1315-1977 | Method of determination of linear density of yarn spun on cotton system. |
| (e) IS:1383-1977 | Method of determination of scouring loss in grey and finished cotton textile materials. |
| (f) IS:1390-1961 | Method of determination of pH value of aqueous extract of textile materials. |
| (g) IS:1544-1973 | Cotton Calico. |
| (h) IS:1670-1970 | Method of determination of Breaking Load and elongation of single strand tenacity of yarn. |
| (i) IS:1720-1978 | Cotton Sewing Threads. |
| (j) IS:1912-1984 | Cotton Yarn Thins. |
| (k) IS:1934-1990 | Method of determination of length and width of fabrics. |
| (l) IS:1963-1969 | Method of determination of threads per unit length in woven fabrics. |
| (m) IS:1964-1970 | Method of determination of weight per square meter and weight per linear meter of fabrics. |
| (n) IS:1969-1968 | Method of determination of breaking load and elongation of woven textile fabrics. |
| (o) IS:2977-1964 | Method of determination (changes in woven fabrics other than wool) on soaking in water. |
| (p) IS:4905-1968 | Method of random sampling. |
| (q) DD/TC/2123 | Laminated Cloth Hessian. |
| (r) CIGS/US/282 | Nylon/Polypropylene Strapping. |

3. STANDARD PATTERN

3.1 The standard pattern of Sheets Hospital Draw held in the custody of Controller, Controllerate of Quality Assurance Textiles & Clothing, Post Box No. 294, Kanpur shall constitute the standard as regard workmanship, finish and any particulars or properties not noted/defined in this specification.

4. MATERIALS

4.1 The following materials shall be used in the manufacture of Hospital Draw.

Materials

Particulars

- (a) Cloth Cotton Calico White
(b) Sewing Cotton 100d Tex X 6 White

IS:1544-1973
IS:1720-1978(V.No.31)

5. MANUFACTURE

5.1 Sheet Hospital Draw is made from Cloth Calico Cotton White in two layers. Thus each sheet shall be in double thickness of material using one piece, creased at one end along the width of the sheet with selvages on either side. The raw edges on the other end shall be turned in 15mm. The two layers of material shall then be sewn together on all sides with two row of machine stitching one on the edges and other 15mm from the edges as shown in the drawing attached to this specification.

5.2 The sheets shall be stitched through out by lock stitch. All ends shall be securely fastened off.

6. REQUIREMENTS

6.1 The Sheets Hospital Draw shall conform to the following dimensions:

(a) Length... 182cm + 2cm - IS:1954-1969

(b) Width ... 91cm + 1cm - IS:1954-1969

(c) The Sheets Hospital Draw shall be stitched with stitches regulated at 35 to 40 stitches per decimeter.

7. MARKING OF STORES

7.1 The sheets, when offered for quality assurance, shall have a white cloth label of suitable size securely caught by inserting in between the selvages at one corner. The label shall be legibly and indelibly marked by ink marking textile black (Cat no. H-1 (a) 8010-000285) with the following details:-

- (a) Cat.No. and nomenclature of store.
(b) Name, initials or recognised trade mark of the manufacturer.
(c) Year of manufacture.

8. SAMPLE PLAN AND CRITERIA OF CONFORMITY

8.1 As QAI No. CQATC/QAI/TC-21/4

9. ACCEPTANCE MARK

9.1 The acceptance mark be affixed on the corner of sheets near size label, using brass acceptance mark of size 2.5cm with indelible ink.

10. PACKING AND MARKING

10.1 Materials (21/01/1984)

10.1.1. The following materials shall be used for packing of Sheets Hospital Draw:-

Materials

- (a) Laminated Hessian Cloth
(b) Twine Jute B.Ply
(c) Polypropylene Strapping
OR
Steel Strapping 16mm wide and 0.9mm thick
(d) Slats of Wood

Specification

IND/TC/2123
IS:1912-1984
GGS/US/282
IS:1029-1970(Med grade)

Best trade quality duly treated with creosote.

treated with creosote

10.2 Method

10.2.1 The store shall be supplied in a new clean and good condition duly preserved.

10.2.2 The Sheets Hospital Draw, suitably folded, shall be placed one over the above. Ten numbers of sheets shall be made into a unit pack by tying them with Twine Jute 3 Ply.

10.2.3 Ten such units packs shall then be wrapped with double layer of Laminated Cloth Hessian to form a compact bale of rectangular shape as far as possible. The overlaps of the inner and outer layers shall be at least 25cm to ensure full protection to the contents of the bale and shall be such that it can be properly and securely seen.

10.2.4 The bales shall be stitched with double 3 ply Jute Twine with not less than 6 stitches per 10cm taking care not to pierce the inner wrapping during stitching.

10.2.5 Sufficient laminated hessian cloth shall be pulled out at each corner to form 'ears' of about 15cm in length.

10.2.6 Slats wood or bamboo strips of approximately 7cm wide and 1cm thick and free from moth or other harmful fungal or insect attack or duly preserved shall be placed lengthwise along the edges of the bale and finally bound with atleast two/three bands of polypropylene strapping or mild steel hoops in each direction along length and width of the bale. The ends shall be sealed with machine.

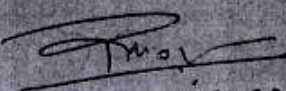
10.2.7 The bale shall be fully pressed and made compact as possible.

10.3 Marking of packages

10.3.1 Before despatch each bale shall be legibly and indelibly marked showing the following details:-

- (a) Nomenclature & Cat No. of the store.
- (b) Quantity packed.
- (c) Serial No. of individual bales and total numbers of bales in the consignment.
- (d) Month and year of packing.
- (e) Name and address of the consignee.
- (f) Gross mass in kg.
- (g) Name or trade mark of the supplier.
- (h) AI/SD No. and date.
- (j) I/Note No. and date.
- (k) Acceptance mark with stencil stamp of size 2.5cm minimum.

Place: KANPUR
Date of Sealing: 3.5.72
Date of Re-sealing: 19 May '92
Date of Issue: 8.10.93


(RAJIV MISHRA)
81x12, F/m
CONTROLLER
CONTROLLERATE OF QUALITY ASSURANCE
TEXTILES & CLOTHING, P.B. NO. 294
KANPUR-208 004

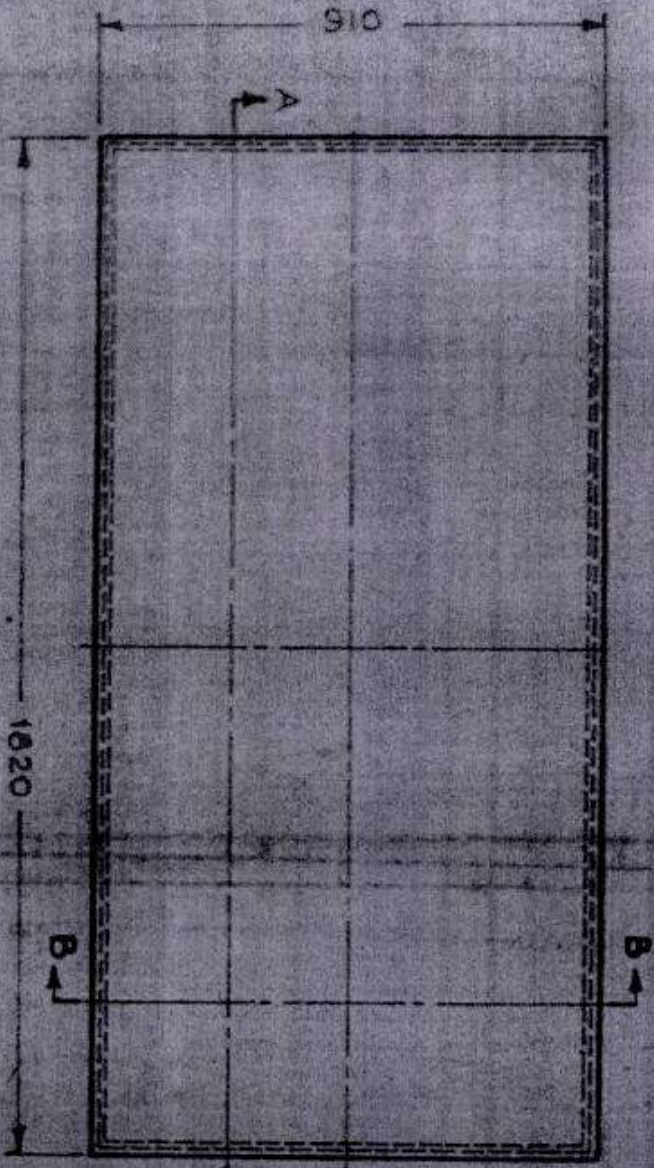
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ORG. ATTACHED TO SPECN. NO. IND/TC/4584 (c)

CAT. NO. K3/7210-000023

SHEETS, HOSPITAL. DRAW

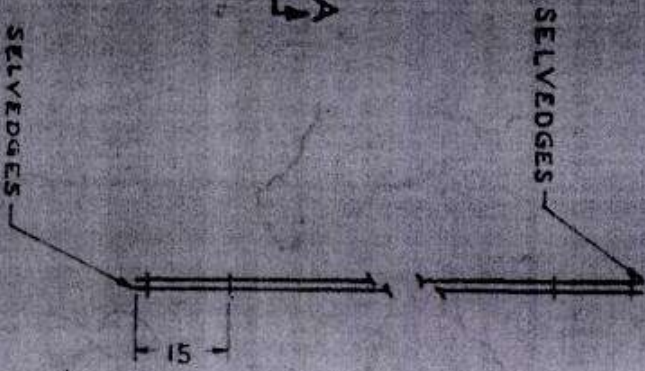
182 - CM. X 91 - CM.



SECTION AA



SECTION BB



MASTER COPY
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IND/TC/4269. (b)

Correct copy of sealed specification at this date.

Kanpur; for CHIEF SUPERINTENDENT DEVELOPMENT TEXTILES & CLOTHING,
Dated... TECHNICAL DEVELOPMENT ESTABLISHMENT TEXTILES & CLOTHING.
for C. I. Chief Inspector of
COA (T & C)
KANPUR
(Based on Home Specification Nos. U/1315 & U/1316)
for CONTROLLER
COA (T & C)
KANPUR

THIS SPECIFICATION or any Patterns, Drawings or other information issued in connection therewith MAY ONLY BE USED for a specific order placed by COMPETENT AUTHORITY; IT IS NOT TO BE USED for any other purpose whatsoever without THE EXPRESS WRITTEN SANCTION OF THE MASTER GENERAL OF THE ORDNANCE IN INDIA.

(Section CE.)

Cat. No. CE 0507
Cat. No. CE 0508

Gowns, Operating, Short.
Sleeves, detachable (for Gowns, Operating, Short).

SPECIFICATION TO GOVERN MANUFACTURE AND INSPECTION.

To be read in conjunction with General Conditions of Contract W.S.B.-87.

REGD.
11/7/7/MC/COI-9

Approved: 1-5-1945.

1. GENERAL.- The garments shall conform in all respects with the sealed patterns deposited with the Chief Superintendent Development Textiles & Clothing, Technical Development Establishment, Textiles & Clothing, Kanpur, and with this specification.

P.T.O.

2. MATERIALS.- The garments shall be manufactured from the following materials and garniture.

1. Drill ^{opaline green} 26/28" 66/71
2. Sewing, Cotton ^{opaline green} 4/60s-White. 9-8 Top 6 opaline green 167
3. Tape Cotton 1/2" (for Sleeves only)
4. Tape Drawstring 1" (for Gowns only).

→ 13mm 2.5mm opaline green

3. MANUFACTURE.- The garments shall conform to the following particulars and size scale :-

All seams, hems, etc. shall be sewn in the same manner as those in the respective scaled patterns. Machine stitching shall be not less than nine or more than eleven stitches to the inch. The size scale should generally be followed. As, however, it is not possible when making garments in wholesale quantities to adhere strictly to the measurement, the following working margins are permissible :-

Gowns:- Breast ^{13mm} 1/2"; Bottom ^{13mm} 1/2"; Length ^{13mm} 1/2"; Half Sleeve 1/4"; Collar 1/4".

Sleeves detachable:- Length ^{6mm} 1/4"; Width ^{6mm} 1/4".

No deviations beyond these margins will be permitted

4. INSPECTION.- If, on examination of ~~any sample of~~ any portion of a consignment, the material is found not to be fully in accordance with this specification, the whole supply may be rejected.

(ii) If, on examination of 20 per cent. of any delivery, 20 percent. of those examined are found not to conform with this specification in pattern, dimensions, workmanship and finish, the whole supply may be rejected.

(iii) All articles not fully in accordance with this specification shall be rejected.

5. MARKING.- Each garment shall ^{1cm} be neatly and legibly marked in permanent ink on the inside in ^{1cm} characters with the contractor's name, initials or recognized trade mark and the year of manufacture thus:-

S - 1945.

6. DELIVERY.- The garments shall be delivered for inspection in a dry and clean condition properly folded, and in bundles of ten. In the case of sleeves detachable in bundles of ten pairs.

SI 233 600 1 L B
GOWNS, OPERATING SHORT.

[illegible]

(1) Including underlap or stand on left side.
Slaves, Dataphase (for Gowns Operating Short)

[illegible]

PG-3-8-51

Cat. No. CE/8415-000013

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GOWNS OPERATING

SIZE SCALE NO. TC/CL0/98147-5

1	2	3	4	5	6	7	8	9
Round gown from edge to edge Breast Bottom		Length of Gown excluding Collar	Half Width of back	Sleeve Length from centre of back	Sleeve Width of Elbow	Width of Hand	Collar Length	Collar Depth
150	165	137	25.5	92.5	46	33	46	2.5
Tolerances		± 1.3	± 1.3	± 1.3	$\pm .6$	$\pm .6$	$\pm .6$	

SCHEDULE OF MATERIALS

Materials

Calico, Cotton, White
Tape, Newar Undyed 12.5 mm.
Sewing Cotton 9.8 Tex/6 White
Buttons, Cotton unfreeable 13 mm.

Particulars

15:1544
To separate IND/TC Specn. 15/1095
15:1720
As per approved sample held by GINCO Kanpur
IND/TC 1206

Packing and Marking

As per Specn. Attached

KANPUR

DATED

23/3/73

For Part 3015
For Part 3015
For Part 3015

29/9/73

WORKING COPY

ANNEXURE TO SIZE SCALE NO. TC/CLO/98 (f)

MARKING AND PACKING FOR GOWNS OPERATING

1. MARKING

1.1 The Gowns operating shall be provided with a cloth label of suitable size legibly and indelibly marked with ink marking textile Black cat No. H-1(a)-8010-000265 (conforming to IS:394) with manufacturers name, initials or recognised trade mark together with year of manufacture, brief nomenclature, cat No. and size of the store. The label shall be securely sewn on the neckband of the Gown.

2. PACKAGING

2.1 The following materials shall be used in packaging :-

<u>SPECIFIED PACKING MATERIAL</u>	<u>SPECIFICATION NO</u>
(i) Polythene film 0.04 mm thick	IS: 2500
(ii) Twine Jute 3 ply	IS: 1912
(iii) Cloth 'C' Heavy	IS: 3781
(iv) Slat wood 0.38mm	
(v) Mild steel strapping 10 mm x	Best trade quality to be previously approved by Inspector
(vi) Mild steel wire 1.8 to 2.0 mm dia	

2.2 The gowns shall be packed as per details given below:

2.3 The gowns shall be delivered/packed in new clean & dry condition.

2.4 10 gowns shall be made into unit pack (bundle) by suitably folding, placing one over the other and then compactly tying them with 3 ply twine jute.

2.5 Six unit packs (bundles) shall then be wrapped with an inner layer of polythene film 0.04 mm thickness and an outer layer of cloth 'C' Heavy or equivalent Hessian cloth to form a compact bale of rectangular shape. The overlaps of the inner layer shall be at least 10 cm to ensure full protection to the contents of the bale. The overlaps of the outer layer of the Hessian cloth shall be such that it can be properly and securely sewn around the bale. The bale shall be stitched with double 3 ply jute Twine with not less than six stitches per dm taking care not to pierce the inner wrapping during stitching. Sufficient hessian cloth shall be pulled out at each corner to form ears of about 15 cm length.

2.6 Slat of wood approximately 7 cm wide and 1 cm thick and free from rot or other harmful fungal or insect attack or alternatively bamboo strips duly preserved shall be placed lengthwise along the edges of the bale and finally bound with at least two bands of metal hoops or alternatively metal wires

in each direction around the length and width of the bale. Each band of metal hoops shall be one complete piece and the ends shall be sealed with the machine or alternatively securely rivetted. Where banding machines ~~are~~ are not available the bales shall properly and securely be bound with metal wires.

2.7 The bales shall be bound with additional bands of metal hoop/wires where the dimensions of the bales may exceed 60 cms in any of the directions, so that no two bands/hoops are more than 25 cm apart.

2.8 In case of ordnance factories wiring of the bales described under sub paras 2.6 and 2.7 above are not applicable.

2.9 In all cases the bales shall be of gross mass of approximately 40 kg each.

2.10 The packing materials mentioned above shall conform to respective specifications.

2.11 The seams of the bales shall be demerred with indelible ink/paint.

3. If ordered for a delivery to a local Inspection Depot the store shall be delivered loose in ~~the~~ firms returnable packs. After inspection the accepted supplies shall be packed by Inspection Depot authorities as described in sub paras above.

4. Before despatch each bale/package shall be legibly marked by stencil showing the following details :-

- (a) Nomenclature and cat no. of the store.
- (b) Quantity packed in the bale
- (c) Lot and serial No. of the bale.
- (d) Month and year of packing
- (e) Gross mass of the bale in kg.
- (f) Name and address of the ~~manufact~~ consignee.
- (g) Inspection Note and date.
- (h) A/T or extract No. and date.
- (i) Name and trade mark of the supplier.

Skg/-

DCV/PROV/BLANKET/2007

INTEGRATED HEADQUARTERS OF MINISTRY OF DEFENCE (NAVY)

INDIAN NAVY SPECIFICATION

ON

BLANKET NAVY BLUE

AND

BLANKET HOSPITAL
(DS CAT NO. 7210-0000007
7210-0000008
7210-0000009

ISSUE DATE: 24 Oct 2007

Directorate of Clothing and Victualling
D-II Wing, Sana Bhawan
Nirman Bhawan P.O.
New Delhi - 110 011

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0. **FOREWORD**

- 0.1 This specification has been prepared by Directorate of Clothing and Victualling on the authority of the Controller of Logistics, Indian Navy.
- 0.2 This specification is for use by the Indian Navy.
- 0.3 This specification has been prepared on the basis of commercially available similar products available with reputed Indian manufacturers and exporters.
- 0.4 This specification would be used for manufacture, quality assurance and procurement of the item.
- 0.5 AHSP for the item covered in this specification is PDCV, New Delhi, for the Indian Navy.
- 0.6 This specification holds good only for the supply order for which it is issued.
- 0.7 The PDCV reserves the right to amend or modify this specification as and when necessary.

1. **SCOPE**

- 1.1 This specification covers the requirements and method of tests for blanket for issue to entitled personnel of Indian Navy as per laid down scale.

RELATED SPECIFICATIONS

2.1 Reference is made in this specification to: -

IS 539:1974	Naphthalene(Second revision)(amendment 1) reaffirmed 1993
IS 744:1977	Method for determination of fibre diameter-projection microscope method(second revision) reaffirmed 1987
IS 1390:1983	Method for determination of pH value of aqueous extract of textile materials(first revision) reaffirmed 1993
IS 1954:1990	Method for determination of length and width of fabrics(second rev)
IS 1963:1981	Method for determination of threads per unit length in woven fabrics(second revision) reaffirmed 1993
IS 1964:1970	Method for determination of weight per square meter and weight per linear meter of fabrics(first revision) reaffirmed 1988
IS 1969:1985	Method for determination of breaking load and elongation of woven textile fabrics(second revision) reaffirmed 1993
IS 2508:1984	Low density polyethylene films(second revision)(amendment 1) reaffirmed 1991
IS 4905:1968	Method for random sampling(amendment 1) reaffirmed 1991
IS 5910:1977	Fineness grade of wool
IND/TC/2123(b)	Laminated cloth Hessian
IND/GS/1683	Polypropylene strapping, 12 mm x 0.5 mm thick
IS 1029(medium grade)	Hot rolled steel Strips(balling) 13 mm wide x 0.4 mm thick

2.2 Copies of Indian Standards are obtainable on payment from: -

Bureau of Indian Standards
Manak Bhawan
9, Bahadur shah zafar marg
New Delhi - 110 002

Or
Their regional/branch offices

2.3 Copies of IND/TC specifications are obtainable on payment from: -

The Controller
Controllerate of Quality Assurance (Textiles & Clothing)
Post Box No. 294, Kanpur - 208 004

2.4 Copies of IND/GS and CIGS/US series of specifications are obtainable on payment from: -

The Controller
Controllerate of Quality Assurance (GS)
Post Box No. 127, Kanpur - 208 001

3. STANDARD PATTERN

3.1 The standard pattern of blanket held in the custody of the PDCV, Integrated Headquarters of Ministry of Defence (Navy), Sena Bhawan, New Delhi - 110011 shall constitute the standard as regards any particulars or properties not noted/defined in this specification.

4. MATERIAL

4.1 The following materials shall be used for manufacture of the blanket: -

(a) Fibre Top

- (i) Fineness grade of wool Top: - 64^s conforming to IS 5910:1977 subject to the condition that the maximum average diameter and minimum average length of the fibre are 22.04 micron and 50 mm respectively.
- (ii) Wool fibre(wool top) used shall be virgin(pure new).
- (iii) The basic blanket shall have a minimum of 80% wool content as per properties detailed above.
- (iv) A maximum of 20% Polyamide fibre of compatible fineness grade is permissible. No other type of fibre is permitted to be used.
- (v) The fibre tops shall be preferably dyed to the required colour and shade prior to blending and subsequent production of yarn. In case piece dyeing is used, care is to be taken to ensure total consistency and uniformity of colour and shade in the end product.
- (vi) Only AZO free dyestuff shall be used.

(b) Piping. The blanket shall have continuous piping of 3.5 cm width on all four sides. The piping shall be of black colour made of 100% high quality polyester satin fabric. The piping shall be sewn to the edges of the blanket with 20 denier nylon(polyamide) yarn of matching colour. The colour of hospital blanket may be red brick or brown as required.

(c) Certification. The blanket shall bear - "WOOL BLEND" certification and logo as authorised by the IWS (International Wool Secretariat).

5. MANUFACTURE

5.1 The count, weave and type of finish of blanket shall be as follows: -

- (a) Count of yarn, universal count(new metric count)
 - (i) Warp - 240 Tex
 - (ii) Weft - 240 Tex
- (b) Weave - 2/2 twill
- (c) Type of finish - Milled and raised

6. WORKMANSHIP AND FINISH

6.1 The blanket shall be properly washed and shall be free from grease, soap, filling or any other admixture which would give fictitious mass or firmness to the finished blanket and from loose surface fibres or specks of any kind.

6.2 In appearance, feel, general workmanship and finish and in all other respects not noted/defined in this specification the blanket shall conform to the corresponding sealed patterns held in the custody of the Principal Director Clothing & Victualling, Integrated Headquarters of Ministry of Defence (Navy), Sena Bhawan, New Delhi - 110 011.

7. MARKING

7.1 Prior to being offered for inspection each blanket shall be affixed with woven satin label of size 15 x 15 cm machine stitched all around with polyester thread of matching colour at one corner duly printed with the following particulars: -

- (a) Section/Ref No. & Nomenclature of the Store.
- (b) Manufacturer's Name, initials or his recognised trademark.
- (c) Month & year of manufacture.
- (d) "Wool Blend" Logo
- (e) To be drycleaned

8. FREEDOM FROM DEFECTS

8.1 The blanket when visually examined both against light and on a surface shall not have more than one objectionable flaw per 8 m length. The objectionable flaws shall be those which strike immediately the eyes of the person examining the blanket and shall be deemed to include: -

- (a) Missing ends and picks.
- (b) Floats.
- (c) Cuts and holes.
- (d) Stains.
- (e) Weft bars and warp section marks.
- (f) Big slubs, knots and specks.
- (g) Dyeing defects(streaks, patches etc).
- (h) Thick and thin places.

8.2 The details of these defects are given in IS: 4125.

9. MOTH PROOFING

9.1 The blankets shall be rendered mothproof by using Naphthalene balls conforming to IS: 539 in suitable quantity (approximately 50 gms per Blanket) or as previously approved by the Quality Assurance Officer.

10. QUALITY ASSURANCE

10.1 Examination of sample taken from any portion of the consignment shall show that the blankets conform to the following requirements when tested in accordance with the method mentioned against each: -

SNo	Characteristics	Requirement	Tolerance	Conforming
(A)	Composition of wool, percent	80	±2	Appendix 'A'
(B)	Composition of polyamide, percent	20	±2	
(C)	Count of yarn, warp	240 Tex	+5%	
(D)	Count of yarn, weft	240 Tex	+5%	
(E)	Breaking strength of single yarn, gms	550	Min	
(F)	Ends per decimeter	80	Min	IS:1963
(G)	Picks per decimeter	60	Min	IS:1963
(H)	Mass per square meter	760 gms	Min	IS:1964
(J)	Mass per blanket under standard atmospheric conditions, Kg	2.7	+0.2 -0.05	
(K)	Breaking load on 15 x 20 cm strips between grips, N.Min			
	(a) Warp	1000	Min	IS:1969
	(b) Weft	700	Min	IS:1969
(L)	Length of the Blanket, cm	230	-5.0 max	IS:1954
(M)	Width of the Blanket, cm	150	-5.0 max	IS:1954
(N)	Average thickness of the finished Blanket under standard condition, mm	2.8	Min	
(P)	Pilling rating	3 or better	Min	
(Q)	Relaxation shrinkage, percent max	5	Max	IS:665(App 'B')
(R)	Loss in mass percent, max	4	Max	IS:2360
(S)	Fineness of wool:	64 ^μ		IS:5010-1977
	(a) Average wool fibre dia microns	38	Max or better	
	(b) Percent fibre distribution: -			
	60 microns & below	90	Min	IS:10014(pt 2):1981
	above 80 microns	5	max	
(T)	Colour fastness to -			
	(a) Light	4 or better		
	(b) Washing	4 or better		
	(i) Staining on adjacent fabric	4 or better		
	(ii) Change in colour	4 or better		
(U)	PH value of aqueous extract	5.0-7.5		IS:1390 cold method

SNO	Characteristics	Requirement	Tolerance	Conforming
(V)	Polyester satin piping(Ribbon) for the			
	(a) Ends per dm	872		
	(b) Picks, per dm	336		
	(c) Wt/linear mtr gms	85		
	(d) Weave	Satin		
	(e) Composition	100% polyester		
	(f) Stitches dm	20		
	(g) Thread to be used	Coats No 6230		
	(h) Colour of thread	Matching colour		
	(i) Linear density of thread	20 denier		
	(k) Type of thread	Filament	Normal	

10.2 Pilot sample: The pilot samples submitted by the manufacturer will be thoroughly evaluated by the Technical Evaluation Committee who in turn will advise the manufacturer in respect of improvement to be made on the end product. Such advice could cover aspects of aesthetics not explicitly covered in the specification but are implicit in a quality product, as well as those of quantified quality parameters laid down in the specifications. The pilot samples could comprise important basic materials as well as the intended end product, at the discretion of the Technical Evaluation Committee, details of which the manufacturer is to ascertain in consultation with the Technical Evaluation Committee well in advance. If considered necessary, the manufacturer may even be advised to submit a fresh pilot sample. A maximum of three pilot samples are permitted to be submitted. If the manufacturer is unable to produce a set of passable pilot samples even in the third attempt, the supply order is liable to be cancelled with necessary encashment/forfeiture of bank guarantee/security as per rules governing the contract. It is to be noted that getting the pilot sample cleared in all respect within the period stipulated in the supply order is the responsibility of the manufacturer, and the therefore the manufacturer is advised to work out his own time schedule with due consideration of a risk of failure in the first, second or even the third and final attempt. The usual time taken by the Technical Evaluation Committee for assessment of a set of samples is about four weeks.

10.3 Pilot sample clearance. On final clearance of the pilot sample, one set will be retained by the Technical Evaluation Committee for future comparison during currency of the supply order and an identical set duly sealed will be handed over to the manufacturer to aid his production as per agreed norms.

10.4 Bulk production clearance. Once a set of samples has been found satisfactory on all counts, the Technical Evaluation Committee will issue bulk production clearance to the manufacturer under intimation to the procurement agency. Thereafter, product has to conform to the approved sample in all respects notwithstanding any specific parameter of this specification having exceeded in the pilot sample.

11. SAMPLING

11.1 Sampling plan shall be as per table below: -

<u>Lot size in Nos</u>	<u>Visual Parameters</u>	<u>Physical Lab Parameters</u>	<u>Chemical Lab Parameters</u>
1 to 50	50	5	3
51 to 100	15	3	3
101 to 150	25	3	3
151 to 300	30	4	3
301 to 500	35	5	3
501 to 1000	40	7	4
1001 to 3000	65	10	5
3001 to 10000	100	15	7

Note: - (i) Sampling shall be done as per IS 4905.

(ii) The inspecting officer shall first draw samples for visual examination as per table above and quality assurance requirement will be prepared by him.

(iii) If found satisfactory in visual examination, the inspecting officer shall draw the samples for lab tests as per above scale for physical parameters, or at any reduced scales as deemed fit.

(iv) Samples for chemical tests and sub samples size will be a part of samples drawn for physical tests.

(v) The test parameters for sub sample size shall be for either certain minor or time consuming lab tests.

(vi) The recommended test parameters for sub sample size shall be threads weave.

11.2 The supplier shall offer the stores serially numbered and arranged in such a manner that it is easily accessible.

11.3 The maximum lot size shall be 5000 Nos.

12. CRITERIA FOR CONFORMITY

12.1 The lot shall be considered to be in conformity with the required standard if the samples drawn for lab test are found satisfactory and the lot is also found otherwise satisfactory in regards to visual parameters as compared to the approved pilot sample. Non conforming lots are liable for rejection in toto.

13. WARRANTY

13.1 The stores supplied against this specification shall be deemed to bear the warranty of the contractor against defective design, material, workmanship and performance for a period of 12 months from the date of receipt of stores at consignee's depot. If during this period, the stores supplied are found to be so defective, the same shall be replaced immediately with the serviceable stores by the contractor at site, free of any charges or cost. Alternatively, the contractor supplying such defective stores shall accept a suitable price penalty for the defective stores as may be decided by the purchasing officer on the recommendation of the consignee/inspecting officer.

14. PACKAGING

14.1 The material used for packing of Blanket shall conform to: -

SNo	MATERIALS	CONFORMING TO
A	Naphthalene	IS:539
B	Jute, twine, 3 ply or polypropylene twine	IS:1912/IS:12734
C	Laminated cloth Hessian	Prov Specn No. IND/TC/2123
D	Polypropylene strapping 12 mm wide x 0.5 mm Thick(or) hot rolled steel strips(balling), 13 mm wide x 0.5 mm thick	IS:1029(Medium grade)
E	Polythene film 0.04 mm thick	IS:2508
F	Cardboard cartons	Best Commercial Quality
G	Self sealing Poly bags	Best Commercial Quality
H	Printed adhesive	Best Commercial Quality

14.2 QUALITY STICKERS

14.2.1 The manufacturer shall provide stickers of approximately size 50 mm x 25 mm free of any additional charges which are to be attached to each unit pack of blanket by the inspecting officer after duly affixing his identification stamp and signature. The inspecting officer will provide samples of stickers to the manufacturer.

14.3 METHOD

14.3.1 The stores shall be delivered/packed in new dry and clean condition.

14.3.2 The blankets shall be free from dirt, dust and other foreign impurities before they are preserved.

14.2.3 The naphthalene shall be evenly distributed/dusted on the blankets and the quantity of naphthalene used for preservation shall not be less than 50 gm per blanket.

14.2.4 When the blankets have been rendered mothproof as above they will be individually folded suitably and packed in unit packages with self-sealing Polybags which will be further sealed.

14.2.5 Twenty unit packages of blankets shall be placed one over the other and then suitably tied with jute twine 3 ply or polypropylene twine. The bundles so formed shall be wrapped with a layer of polythene film and placed in a rectangular cardboard carton of suitable size. The carton, thereafter will be sealed with adhesive tapes and bound with polypropylene tapes. The roll of adhesive tapes are to bear the manufacture's identification marking duly printed on its entire length. The carton shall further bear its gross weight as well as other suitable identification marking neatly printed/stenciled on it. Thereafter, the carton shall be covered with an outer layer of laminate cloth hessain stitched all around with double twine 3 ply jute or polypropylene with not less than 6 stitches per 10 cm, taking care not to pierce the cardboard carton itself. Various identification marking will thereafter be again stenciled on the exposed surface of the hessain cloth. The bulk package, thus formed will be suitably tape bound with minimum two each longitudinal and transverse polypropylene straps and then finally sealed.

14.2.6 The seams on the hessain cloth shall be tamper-proof with indelible ink/paint.

Note: - (i) The packages shall be fully pressed and made compact as much as possible.

(ii) In case of non-availability of specified/standard packing materials other alternative materials mentioned above may be used as substitute only with prior permission of inspecting officer.

14.4 **MARKING**

14.4.1 Before despatch each bulk package shall be legibly/indelibly marked by stencil showing the following details: -

- (a) Nomenclature and DS cat number of stores.
- (b) Quantity packed in the package.
- (c) Batch number and date.
- (d) Month and year of packing.
- (e) Manufacturer's name initials or recognised trademark.
- (f) Gross mass of package in kg.
- (g) Inspection note number and date.
- (h) Contract number and date.
- (i) Name and address of the consignee.

15. DEFENCE STORES CATALOGUE NUMBER

15.1 The store covered by this specification bear DS Cat No. N7210-000007 to 9.

16. SUGGESTIONS FOR IMPROVEMENT

16.1 Any suggestion for improvement in this document may be forwarded to: -

The Principal Director
Integrated Headquarters of Ministry of Defence (Navy)
Directorate of Clothing & Victualling
Sena Bhawan
New Delhi - 110 011

APPENDIX 'A'

METHOD FOR DETERMINATION OF COMPOSITION

A.1 The percentage of wool in the blanket shall be determined as per method given below:

A.1.1 Take the specimen weighing approx 5g and dissect the same into yarn and cut the yarn into pieces of approx 25 mm length.

A.1.2 Dry the pieces of the test specimen at 105° to 110° C to constant mass and determine its oven dry mass.

A.1.3 Put all the pieces of the test specimen in a suitable conical flask/beaker. Add 5 percent sodium hydroxide solution to make a material to liquor ratio of 1:100. Boil the contents of the flask/beaker for 10 to 15 minutes.

A.1.4 Decant the liquid through a weighted sintered glass filter crucible with application of necessary suction of complete filtration and again with water until the water is neutral to litmus.

A.1.5 Dry the residue at 105° to 110° C to constant mass and determine its oven dry mass.

A.2 CALCULATIONS

A.2.1 Calculate the percentage by mass of wool (x) in the test specimen by the formula given below: -

$$\text{Percentage of wool fibre} = \frac{m - m_1 \times 100}{m}$$

Where

m - Oven dry mass of the specimen

METHOD FOR DETERMINATION OF RELAXATION SHRINKAGE

1. A sample test piece of approx 60 x 60 cm shall be cut (not torn). Accurate markings of 50 cm shall be made in warp and weft direction and then the test piece shall be immersed in cold water for 2 hrs taken out, laid flat and allowed to dry on a flat smooth surface in an atmosphere of 65 $\pm$ 2 percent RH and 27°C $\pm$ 2 degree C temperature. Measurements shall be taken under plate glass before and after the treatment and the percentage of contraction be calculated in both the directions. Three determinations shall be made on each test specimen in each direction.

DCV/PROV/BLANKET/2007
BLANKET NAVY BLUE/BRICK RED/BROWN

INTEGRATED HEADQUARTERS OF MINISTRY OF DEFENCE (NAVY)
DIRECTORATE OF CLOTHING & VICTUALLING

AMMENDMENT NO 1 - INDIAN NAVY SPECIFICATION

ON

BLANKET NAVY BLUE/BRICK RED/BROWN

AND

BLANKET HOSPITAL

DS CAT NO. 7210000007 - 9



ISSUED DATE: 14 JAN 2020

DIRECTORATE OF CLOTHING AND VICTUALLING
INTEGRATED HEADQUARTERS, MINISTRY OF DEFENCE (NAVY)
ROOM NO 103, TALKATORA ANNEXE
TALKATORA STADIUM
NEW DELHI - 110 001


Commandant
Directorate of Clothing & Victualling
Integrated Headquarters, Ministry of Defence (Navy)
Talkatora Stadium, New Delhi - 110 001

RECORD OF AMENDMENT

[illegible]

Omde Gurnant Singh
Omde
D/o of Clothing & Virtualising
Intelligence HIC Model (New)

Amendment – 1

4. Material

4.1 The following materials shall be used for manufacture of the blanket:-

(b) Piping.

For : The piping shall be of Black colour made of 100% high quality polyester satin fabric.

Read : The piping shall be made of 100% high quality polyester satin fabric with matching colours as defined in this specification hereafter.

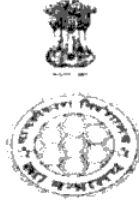
- (i) The piping shall be of Navy Blue colour for Blanket Navy Blue.
- (ii) The piping shall be of Brick - Red colour in case of Blanket Brick Red.
- (iii) The piping shall be of Brown colour in case of Blanket brown.

[illegible]

Towel operating

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JSS 7210 - 08 : 2013
(Revision No. 1)



रक्षा मंत्रालय
MINISTRY OF DEFENCE

ज्वाइंट सर्विस स्पेसिफिकेशन
JOINT SERVICES SPECIFICATION

ON

TOWEL OPERATING, OPALINE GREEN
(DS Cat No. 7210 - 000 065)

मानकीकरण निदेशालय
रक्षा उत्पादन विभाग
रक्षा मंत्रालय
'एच' - ब्लॉक, निर्माण भवन डाकघर
नई दिल्ली - 110 011

DIRECTORATE OF STANDARDISATION
DEPARTMENT OF DEFENCE PRODUCTION
MINISTRY OF DEFENCE
'H' - BLOCK, NIRMAN BHAVAN, P.O.
NEW DELHI - 110 011

LIST OF MEMBERS ASSOCIATED WITH FORMULATION OF THIS STANDARD

1. The revision of this Joint Services Specification has been approved by Shri Sanjeev Kapur Addl DGQA (S), Chairman, Stores Standardisation Sub - Committee.

2. The following members have been associated in preparing the document : -

Sl. No.	Name and Designation	Organisation
(a)	Wg Cdr SKS Gosain, JDQAS (S & C)	DQAS (Aero) / Air HQ
(b)	S. Chakraborty, SSO - I	CQA (T & C)
(c)	Dr. S. K. Pandey, SSO - II	CQA (M)
(d)	Shri C.B. Pandey, JAG (NFSG)	CQA (PP)
(e)	Shri D K Pujari, SSO - I	CQA (GS)
(f)	Cdr. Naveen Kumar (IN)	DQA (N)
(g)	Shri Jarnail Singh, Dy Director	DGAQA
(h)	Sh. Suraj Raj, SSO - II	DGAQA
(j)	Maj. S. Bansal, DD OS (PII)	MGO / CP Cell / OS Dte.
(k)	Shri A K Handa, Jt Dir. Stdn Cell (Army)	MGO Br / EM / GS & C
(l)	Wg Cdr Chaturvedi, Deputy Director	Dte of Stores / Air HQ
(m)	Lt. Col Pankaj Bisht, Jt Dir EM (Std) (Army Std Cell)	MGO / EM / Std
(n)	Shri BL Jatav - NSO	NHQ / DCV
(o)	Cdr R Venkatesan, JDME	NHQ / DOS (E)
(p)	Dr. D N Tripathi, Sc. 'F'	DMSRDE
(q)	Shri S B Yadav, Sc. 'F'	DMSRDE
(r)	Cdr. Manish Singh (JDLS)	JDLS
(s)	Shri S.K. Bandyopadhyay, JGM	OEF, Kanpur
(t)	Shri Ajay Satija, AE (QA)	DSC, Kanpur
(u)	Lt. Col. Parveen Deo	Secy, SSSC

RECORD OF AMENDMENTS

Amendment		Amendment pertains to: Sl.No. / Para No. / Column No.	Authority	Amended by	Signature & Date
No.	Date			Name & Appointment (IN BLOCK LETTERS)	

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0.0 FOREWORD

0.1 This specification has been prepared by Stores Standardisation Sub - Committee on the authority of Standardisation Committee, Ministry of Defence

0.2 This specification has been approved by the Ministry of Defence and is mandatory for use by the Defence Services.

0.3 This Specification is revision of JSS 7210 - 08 : 1981 (Reaffirmed - 2008) and supersedes the same.

0.4 This JSS would be used for Manufacture, Design, Quality Assurance and Procurement of the item.

0.5 Quality Assurance Authorities for the item covered in the specification are CQA (T & C), Knapur, DC & V, New Delhi and DQAS (Aero), New Delhi for Army, Navy and Air Force respectively. Enquiries regarding technical parameters shall be addressed to the Quality Assurance Authority, while other enquiries shall be referred to : -

The Director,
Directorate of Standardisation,
Ministry of Defence,
'H' Block, Nirman Bhavan PO,
New Delhi - 110 011

0.6 Non - registered users can obtain the following on payment : -

(a) Copies of IS from : -

Bureau of Indian Standards,
Manak Bhavan,
9, Bahadur Shah Zafar Marg,
New Delhi - 110 002
or
their regional / branch offices

(b) Copies of JSS / JSG from : -

The Director,
Directorate of Standardisation,
Standardisation Documents Centre
Ministry of Defence
Room No.05, 'J' Block, Nirman Bhavan PO,
New Delhi - 110 011

0.7 Indian Standard (IS) specifications are available free of cost for registered users on : -

Directorate of Standardisation Website
www.defstand.gov.in
for registration visit our website.

0.8 This specification holds good only for the supply order for which it is issued.

0.9 Directorate of Standardisation Website. All the approved JSSs / JSGs are available on the Directorate of Standardisation Website **www.defstand.gov.in** Defence organisations desirous of accessing a copy of this document are requested to approach the Directorate of Standardisation for obtaining user id / password to access the website.

1.0 SCOPE

1.1 This specification covers the requirements of Towel Fabricated from Dosuti.

2.0 RELATED SPECIFICATIONS

2.1 Reference is made in this specification to :-

Sl. No.	Specification / Document No.	Nomenclature
(a)	IS 179 : 2009 (Third Revision) Amd - 5	Textiles - Cotton dosuti - Specification
(b)	IS 1720 : 1978 (First Revision) Reaffirmed - 2012	Specification for Cotton Sewing Threads
(c)	IS 2508 : 1984 (Second Revision) Reaffirmed - 2008 Amd - 3	Specification for Low Density polyethylene Films
(d)	IS 2771 : Part 1 : 1990 (Second Revision) Reaffirmed - 2008 Amd - 3	Corrugated Fibreboard Boxes – Specification : Part 1 : General Requirements
(e)	IS 4905 : 1968 Reaffirmed - 2011 Amd - 1	Methods for random sampling
(f)	IS 9738 : 2003 (Second Revision) Reaffirmed - 2010	Polyethylene bags for general purposes
(g)	IS 13262 : 1992 Reaffirmed - 2008 Amd - 1	Pressure sensitive adhesive tapes with plastic base - Specification
(h)	JSS 8135 - 10 : 2004 (Revision No. 1) Reaffirmed - 2011 Amd - 1	Polypropylene Strapping, 12 mm wide

3.0 STANDARD PATTERN

3.1 The Standard Pattern, held by the Controller, Controllerate of Quality Assurance of (Textiles & Clothing) Kanpur shall constitute the standard as regards any particulars or properties not noted / defined in this specifications.

4.0 MATERIALS

4.1 The Towel shall be made from the following materials conforming to the particulars quoted against each : -

Sl. No.	Components	Materials	Particulars
(a)	Body	Dosuti, Opaline green	IS 179 : 2009
(b)	Tape	Tape, made from Dosuti opaline green	- do -
(c)	Sewings	Sewing Cotton, 9.8 tex / 6, Opaline Green	IS 1720 : 1978

5.0 MANUFACTURE

5.1 Body – The Towel shall be made from a single piece of Dosuti Opaline Green. The two raw ends of the Dosuti shall be hemmed 20 mm with a turn – in of 20 mm and secured with one row of machine stitching as shown in the plate. The selvedge shall not be hemmed. Tape will be manufactured from two fold of Dosuti Opaline Green. Both the raw edges will be turned 5 mm inside and will be stitched together with single row of machine stitched. The width of finished tape will be 13 mm. A 230 mm long piece of tape shall be sewn at the center of one of the hem side of towel to form a loop. The effective length of the loop (on the double) shall be 100 mm as shown in the attached plate.

5.2 Sewings – The Towel shall be stitched throughout with lockstitch regulated at 11 to 13 stitches for every 30 mm.

6.0 DIMENSIONS

6.1 The Towel shall conform to the dimensions shown in the plate attached to this specification. A tolerance of ± 15 mm shall be permitted in the length and width of the towel.

7.0 WORKMANSHIP AND FINISH

7.1 In appearance, shade, general workmanship and finish and in all other respect not noted defined in this specification the towel shall conform to the Standard Pattern.

7.2 During Bulk QA of the store 100 % store to be checked for workmanship, make, shape and finish.

8.0 MARKING

8.1 Each Towel, prior to being offered for inspection shall have a White cloth label of 50 x 40 mm securely sewn to one of the corners on inside of the cloth on which the supplier shall legibly and indelibly mark his name , initials or recognised trade mark, the year of manufacture, brief nomenclature and DS Cat number of the store.

9.0 PRE - INSPECTION OF STORES / CONSIGNMENT

9.1 Manufacturers / contractors must satisfy themselves that the store is in accordance with the terms of the contract and fully conform to the required specification, by carrying out a thorough pre - inspection of each lot before actually tendering the same for inspection to the Quality Assurance Officer nominated under the terms of the contract. A declaration by the contractor that a necessary pre - inspection has been carried out on the stores tendered will be submitted alongwith the challan. The declaration will also indicate the method followed in carrying out pre - inspection showing the features checked/tested and will have the test certificate attached to the challan / declaration.

9.2 If the Quality Assurance Officer finds that the pre - inspection of the consignment as required above has not been carried out, the consignment is liable for rejection.

10.0 SAMPLING

10.1 Formation of Lot

10.1.1 The delivery shall be visually inspected by Quality Assurance Officer on the spot in the first instance to ascertain its homogeneity in respect of nature, source, year of manufacture, batch, uniformity of production etc. If the product units are homogeneous, the delivery shall be treated as one lot.

10.2 The supplier shall arrange the unit of the homogenous lot in such a way that all the product units are easily accessible to the sampling Officers from all sides to enable him to draw samples at random from any portion of the homogeneous lot.

10.3 The sample for visual examination and laboratory test to be drawn from the lot at random as per IS 4905 and No of sample to be drawn from the lot as per table given below.

SAMPLING PLAN FOR MADE UP STORE

Based on 4 % AQL

Authority IS 2859

Lot size in No. Bundles of 5 each	Sampling Plan for					
	Visual Parameters		Physical Parameters (for laboratory tests)		Chemical parameters and other Physical Parameters* requiring long testing time (for laboratory tests)	
	Sample Size	Acceptance No.	Sample Size	Acceptance No.	Sample Size	Acceptance No.
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Bellow – 150	13	1	13	1	3	0
151 – 280	13	1	13	1	5	0
281 – 500	20	2	13	1	5	0
501 – 1200	32	5	20	2	5	0
1201 – 3200	50	5	32	3	13	1

NOTE : (1) Samples drawn as per column (2) to be first examined visually. If found satisfactory, the samples for lab testing as per column (4) be drawn out from the samples originally drawn as per column (2) by the sampling officer and marked by him accordingly.

(2) Samples for lab testing will be drawn from the samples drawn as mentioned in column (2) only.

(3) For chemical parameters samples to be drawn at random as per IS 4905 from those samples drawn as per column (4).

(4) Lot size : 10000 Numbers.

11.0 CRITERIA FOR CONFORMITY

11.1 All the sample units drawn as per clauses above shall be tested / examined to the relevant specification requirements. If all the sample units meet the specification requirements, the lot shall be considered to be in conformity otherwise not.

12.0 WARRANTY

12.1 Except as otherwise provided in the invitation to the tender, the contractor / seller hereby declare that the goods, stores, articles sold / supplied to the purchaser under this contract shall be of the best quality and workmanship and new in all respect and shall be strictly in accordance with the specifications and particulars contained / mentioned in the contract.

The contractor / seller hereby guarantees that the said goods / stores / articles would continue to conform to the description and quality aforesaid for a period of 12 months from the date of delivery of the said goods / stores / articles to the purchaser or 15 months from the date of shipment / dispatch from the contractor's work whichever is earlier and that notwithstanding the fact that the purchaser (Inspector) may have inspected and / or approved the said goods / stores / articles, if during the aforesaid period of 12 / 15 months the said goods / stores / articles, be discovered not to conform to the description and quality aforesaid or not giving satisfactory performance or have deteriorated and the decision of the purchaser in the behalf shall be final and binding on the contractor / seller to rectify / replace by acceptable goods / stores / articles or such portion or portions thereof as is found to be defective by the purchaser within a reasonable period not exceeding 03 months or as may be allowed by the purchaser in his discretion on the application made thereof by the contractor / seller and in such an event the above mentioned warranty period shall apply to the goods / stores / articles rectified / replaced from the date of rectification / replacement thereof, otherwise the contractor / seller shall pay to the purchaser such compensations as determined by the purchaser as may arise by reason of the break of the warranty herein contained.

13.0 PACKING

13.1 The following materials shall be used for packing of sheets of various :-

Sl. No.	Material	Conforming to
(a)	Polythene Film 0.04 mm thick	IS 2508 : 1984
(b)	Double walled corrugated fibre board boxes, 5 ply (outer liner shall be laminated with polyethylene to provide water – proofness)	IS 2771 : Part 1 : 1990
(c)	Polyethylene bags for general purposes	IS 9738 : 2003
(d)	Pressure sensitive adhesive tapes with plastic base	IS 13262 : 1992
(e)	Polypropylene Strapping, 12 mm wide	JSS 8135 - 10 : 2004 (Revision No. 1) Reaffirmed - 2011 Amd - 1
(f)	Label cardboard 45 x 25 mm with tie - on - tag	Best Trade Quality

13.2 Method

13.2.1 The stores shall be supplied in new, clean, dry and sound condition.

13.2.2 Each sheet shall be suitably folded and kept in a polythene bag. 10 such sheets (kept in polythene bag) shall be tied by a suitable twine / cord to form a bundle, suitable No. of such bundles of one size shall be placed in a double walled corrugated fibre board box of suitable size previously lined with polythene film.

13.2.3 The flaps of the boxes shall be closed with pressure sensitive adhesive tape with plastic base shall be affixed fully covering the joints.

13.2.4 The boxes packed above shall be bound with two bands of polypropylene strapping in each direction along length and width. Each strapping shall be one complete piece and the ends shall be sealed with machine using steel seals or by heat sealed.

13.2.5 The gross mass of the box shall not exceed 40 kg.

14.0 DEFENCE STORES CATALOGUE NUMBER

14.1 This stores / item have been allotted following DS Cat Part : -

DS Cat Number	Nomenclature
7220 – 000 065	Towel Operating Opaline Green

15.0 SUGGESTIONS FOR IMPROVEMENT

15.1 Any suggestions for improvement of this document may be forwarded to : -

The Director
Directorate of Standardisation
Ministry of Defence,
'H' Block' Nirman Bhavan PO
New Delhi – 110 011



ADDITIONAL CLAUSE AGAINST GeM BID FOR PROCUREMENT OF HOSPITAL CLOTHING.

1. **Specification & Consignee:** Details of Entitled Clothing Items are as follows.

Ser.	Item Description	Specification Details	Details of Qty (In Nos) per Consignee		
			MO (MBI)	BVY (KAR)	Total
1	BLANKET HOSPITAL BROWN	DCV/PROV/BLANKET/2007 DT. 24 OCT 07	144	0	144
2	TOWEL OPERATING	JSS7210-08:2013	2352	0	2352
3	GOWN OPERATING SIZE M	IND/TC:4269 (B)	269	0	269
4	GOWN OPERATING SIZE L	IND/TC:4269 (B)	167	0	167
5	SAREE COTTON LIGHT BLUE	JSS 810-05:2014	1807	0	1807
6	SHEET DRAW HOSPITAL	IND/TC/4554 (C)	119	0	119
7	SHEET HOSPITAL WHITE BLEACHED 255 CMSX175 CMS	JSS:7210-25:2019	9175	500	9675

2. **Sample.** Firm to submit **01 Nos. of** as sample of tendered item compliant to the specification mentioned in GeM tender enquiry to following address **prior bid end date** quoting reference of the item description, GeM bid number & bid end date. Submit with 03 copies of sample forwarding/submission letter. Samples will not be accepted, if above mentioned details are not mentioned in packages / consignment. The samples to be forwarded on No Cost No Commitment basis.

Address : Asst. Base Logistic Officer (Clothing), 1st Floor, Afloat Clothing Centre - II, RIS Building, Colaba, Old Navy Nagar, Mumbai – 400 005, Tel : 022 – 22147471.

Note : Samples being sought from bidder for evaluation and comparison with tendered specification and to scrutinize the samples with the lab test report submitted by bidders. **Samples will not be evaluated on the basis of feel, fit and finish.**

3. **Lab test Report :** Bidders are required to submit Govt./NABL accredited lab report along with online technical bid on GeM portal. The test report should not older than **06 months**. It is mandatory to get all parameters of the specification tested by the lab. If lab test report of any of the parameters as per specification is missing the technical bid is liable for rejection. Further, lab testing of the item will be carried out at inspection stage / post order placement / post supply of material or at any stage. **Bids without lab test report / incomplete lab test report will be liable for rejection outright.**

Note:- Expenditure for lab test at any stage will be done by the firm.

4.

Embossing / Labelling :-

- 4.1 Firm's Name _____
 4.2 Purchase order reference _____
 4.3 Batch No., Month & year of Manufacturer _____
 4.4 Pattern No. & Nomenclature **CNCMT-** _____ & _____)
 4.5 Instruction for upkeep and maintenance of the item, if any
 4.6 Logo and Property of Indian Navy

5. **L1 Determination:** All firms participating on GeM with valid documents/bids will be considered, thereafter Board of Officers will evaluate technical documents, lab test report and samples submitted by every firm and qualify bids of only those firm which is acceptable on parameters (technical documents, lab test report and sample). CST will be prepared and L1 will be determined by GeM portal.

6. **Delivery Period: - PSPP in 01 Lot**

- (i) Items to be offered for inspection within 140 days from receipt of GeM contract.
 (ii) Estimated time for inspection & issue of inspection note is 30 days.
 (iii) Delivery of item at MSD(Mankhurd) within 10 days of receipt of inspection note
 (iv) Total delivery period will be 180 days (140+30+10).

7. **Inspection:** The inspection will be carried out by Board of Officer/Inspection Agency nominated by the buyer against specifications for tendered items. All expenses towards inspection during pre-dispatch inspection will be borne by the firm, **Undertaking for pre-dispatch inspection to be submit on firm's letter head along with GeM bid.**

8. **Warranty/Guarantee:** The bidders are requested to submit the Warranty / Guarantee certificate of the tendered items for at least 02 years. Any defects / substandard report within 02 years would entail the firm to replace the defective lots within the warrantee period. All participating firms are to submit an undertaking for the same. Suppliers are to submit W/G certificate from the OEM/Manufacturer.

9. Vendor evaluation technical compliance matrix (enclosed) to accepted/filled by respective bidders. **Bids without MATRIX will be liable for rejection outright.**

10. Bidders have to strictly adhere to published time line w.r.t tender opening date and delivery period. **No request for extension will be entertained.** Bidder anticipating insufficient delivery period time line may choose not to participate in the tender.

11. **Bid Security Declaration:-**Refer to DoE Memo Number F.9/4/2020-PPD dated 12 Nov 20 all the prospective firms / bidders to submit 'Bid Security Declaration Certificate' as per performa enclosed. **Bids without this documents will be liable for rejection outright.**

12. The firm will be responsible for unloading of consignment at Consignee's premises.


 (Mandanna KP)
 Lt Commander
 Asst. Base Logistics Officer (Clothing)
 for Base Logistics Officer

BID SECURITY DECLARATION FOR
GEM BID NUMBER GEM/B/2025/

1. Refer to DoE MoF OM No. F.9/4/2020-PPD dated 12 Nov 20.
2. I/We confirm that we accept that, if I/We withdraw or modify our bid during the period of validity etc. we will be suspended for Six Months.

Stamped and Signature of Bidder / Firm

VENDOR EVALUATION COMPLIANCE MATRIX **AGAINST GeM BID**

TO BE SUBMITTED WITH ONLINE TECHNICAL BID ON GEM

<u>GEM RFP CRITERIA</u>	<u>COMPLIANCE STATUS (YES / NO)</u>	<u>SUPPORTING DOCUMENTS</u>	<u>REASON FOR NON COMPLIANCE</u>
Experience certificate / Past purchase order		Certificate / order copies to be attached	
Bidder Turnover for last three (3) years		Audited Profit & Loss and Balance sheet by CA / turnover certificate from CA to be attached on GeM portal.	
Bid security declaration		Copy of bid security declaration on firm's letterhead to be attached on GeM portal.	
Acceptance of item compliance technical matrix		Technical compliance matrix to be accepted / filled by respective bidders to be attached on GeM portal.	
Lab Test Report (As per Specification)		Attached Govt. / NABL accredited test report with online technical bid on GeM portal.	
Undertaking item to be supplied as per tendered specification mentioned in GeM bid.		Copy of Undertaking on firm's letterhead to be attached on GeM portal.	
Submitted 01 Nos. of tendered item to buyer		Copy of proof of sample submission to be attached on GeM portal.	
Undertaking for acceptance of RFP Warranty / Guarantee clause		Copy to be attached on GeM portal for undertaking for acceptance of RFP Warranty / Guarantee clause.	

Note : Bids without above documents will be liable for rejection outright.

Stamped and Signature of Bidder / Firm